IBM SkillsBuild | AICTE | BharatCares (CSRBOX)

Machine Learning & Applied AI Internship Program 2026

Concept Note

AI Shield

A Unified, Explainable Fraud Detection Platform for Scam Messages, Phishing URLs and Emails using Machine Learning

Submitted by	Karthik Siraparapu
Internship ID	IBMUEML004123
College	Marwadi University
Department	B.Tech Computer Science & Engineering (AI & ML), Semester 5
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1. Executive Summary

Fraud that reaches people through SMS, messaging apps, links and e-mail has become one of the most common ways in which digital-payment users lose money and personal data. Rule-based filters struggle with newly worded scams, and most tools return only a yes/no answer that teaches the user nothing. **AI Shield** is a Streamlit web application with three machine-learning modules - a scam-message analyzer, a phishing-URL detector and an e-mail checker - that returns a verdict, a fraud probability and the reasons behind it.

The project's distinguishing feature is its **validation-first approach**. Before training, every supplied dataset was audited. This exposed a randomly labelled 100,000-row dataset (excluded), 17% duplicate rows and 2,900 train/test leaks in the scam-transcript data (removed), 61% duplicate rows in the 525K e-mail feature table, and URL datasets that contain only features, not URLs. On the cleaned, leak-free test set the scam-text model reaches **99.7% accuracy (F1 0.998, ROC-AUC 0.9997)**. The URL models, limited to features that can be computed from a pasted link, reach ROC-AUC 0.81-0.83. The e-mail text model scores F1 0.88 on held-out Enron data but fails on an independent e-mail set (ROC-AUC 0.49); it is therefore shipped as an explicitly experimental module. Reporting these limits is a deliberate design decision.

2. Introduction

India's rapid adoption of UPI, mobile banking and online shopping has created a large population of first-time digital users. Fraudsters exploit this with fake KYC alerts, OTP requests, lottery prizes, courier notices, loan offers and impersonation of banks or police. Detecting such content automatically is a natural binary-classification problem that combines natural language processing, feature engineering on URLs and applied cybersecurity. This note describes the problem, the data, the methodology and the results of AI Shield, built as the final project of the AICTE-IBM SkillsBuild Machine Learning & Applied AI internship.

3. Background

Common attack types addressed by the project:

- **Smishing / scam messages** - SMS or chat messages urging the reader to share an OTP, update KYC, claim a prize or pay a fee.
- **Phishing URLs** - look-alike domains, IP-address hosts, shortened links and long URLs with sensitive keywords such as login or verify.
- **Phishing and fraud e-mail** - password-reset lures, wire-transfer requests and prize notifications.
- **Voice/transcript scams** - 'digital arrest' and impersonation scripts, represented in the scam-transcript dataset.

Machine learning helps because it can learn word patterns and structural signals from labelled examples instead of relying on hand-written rules, and can be retrained as scammers change their wording.

4. Problem Statement

Fraudulent SMS messages, phishing links and e-mails reach users faster than static spam filters can adapt, and users receive no explanation of why a message is dangerous. This project builds an explainable, machine-learning-based fraud detector that classifies a pasted message, URL or e-mail as legitimate or fraudulent, reports a probability and highlights the evidence, while being transparent about the reliability of each model.

5. Need for the Project

- Victims often decide within seconds; a free second-opinion tool can interrupt a scam before money is sent.
- Existing alternatives (operator spam filters, blocklists, helplines) are reactive: they need a prior report, or they cover only known URLs.
- Explanations teach users to recognise scam patterns themselves, improving long-term cyber-awareness.
- Honest validation is needed: many student projects report inflated accuracy because of duplicates and data leakage. AI Shield demonstrates how to avoid this.

6. Literature Review

Research on text-based spam detection established the bag-of-words / TF-IDF representation with linear classifiers as a strong baseline: Joachims (1998) showed that support vector machines handle the very high-dimensional, sparse features of text well, and Almeida et al. (2011) released the widely used SMS Spam Collection and compared classifiers on it. TF-IDF weighting itself follows Salton and Buckley (1988). AI Shield adopts this proven combination for its text modules.

For phishing websites, Mohammad, Thabtah and McCluskey (2012, 2014) defined the UCI phishing feature set (URL length, '@' symbol, sub-domains, prefix/suffix hyphens, IP-address hosts, HTTPS token and others) that is also used in this project. Sahingoz et al. (2019) studied machine-learning detection from URLs and showed that lexical features alone give useful but imperfect separation - consistent with the ROC-AUC of about 0.8 observed here. Random forests (Breiman, 2001) are a robust choice for such tabular features.

The Enron corpus (Klimt and Yang, 2004) is a standard e-mail research dataset; a fraud-labelled version is used here. Because it contains one organisation's mail from a single period, models trained on it risk learning corporate writing style rather than fraud, a domain-shift effect measured in Section 9. Finally, interpretable models and per-prediction explanations (Ribeiro et al., 2016) motivate the explainable-AI layer: for linear models the contribution of each TF-IDF term can be read directly from the learned weights.

7. Proposed AI Solution

AI Shield is a single web application with three modules sharing one interface and one explanation layer:

- **Scam-message analyzer** - cleans the text (HTML removal, URL masking, lower-casing), converts it to word 1-2-gram TF-IDF features and scores it with a calibrated linear SVM.
- **Phishing-URL detector** - computes lexical features from the URL string and averages the probabilities of two Random Forest models trained on two different datasets.
- **E-mail checker (experimental)** - TF-IDF linear SVM on subject plus body, together with structural features (links, domains, urgent keywords).
- **Explainable-AI layer** - an Indian-context fraud lexicon (OTP, KYC, UPI, RBI/SBI impersonation, urgency, prize bait), the terms the model weighted most, and URL red flags. A transparent rule lifts a 'Legitimate' verdict to 'Suspicious' when several independent indicators fire.
- **Threat dashboard** - charts of 3,000 global cyber-incident records (2015-2024) by year, attack type, industry and loss.

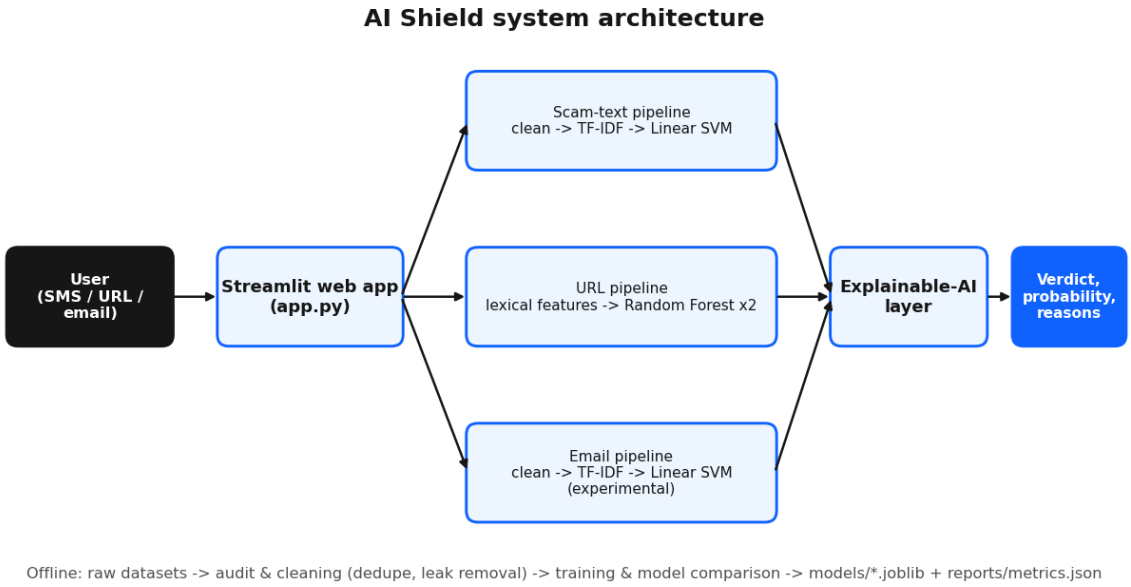


Figure 1. AI Shield architecture.

8. Dataset Description and Data Audit

Ten files were supplied. Each was inspected for size, missing values, duplicates, label distribution and fitness for its task before any model was trained (Table 1, Figure 2).

Dataset	Rows	Finding	Decision
Composite Scam Transcripts	36,751 train / 9,188 test	17% duplicate train rows; 2,900 test texts also in train	Deduplicated, leak removed; used for scam text
GCT 100K	100,000	Labels independent of text and attack type; text-only model 49.9% (chance)	Excluded from training
Phishing URL features (basic)	2,488	46% duplicates; features only, no URL strings	1,354 unique rows; 12 computable features
UCI phishing features	11,055	Features only; many need page content	8 lexical features that a URL string provides
Email features	524,846	61% duplicates; 3% positive	Supplementary experiment only
Enron fraud-labelled	447,417	0.5% fraud; corporate-only negatives	Balanced sample of 10,451 for e-mail text
Email spam (independent)	84	Tiny but modern	External validation only
Global Cyber Threats	3,000	Clean	Dashboard
CryptoScamDB URIs	17	No labels	Demo only

Table 1. Dataset audit summary.

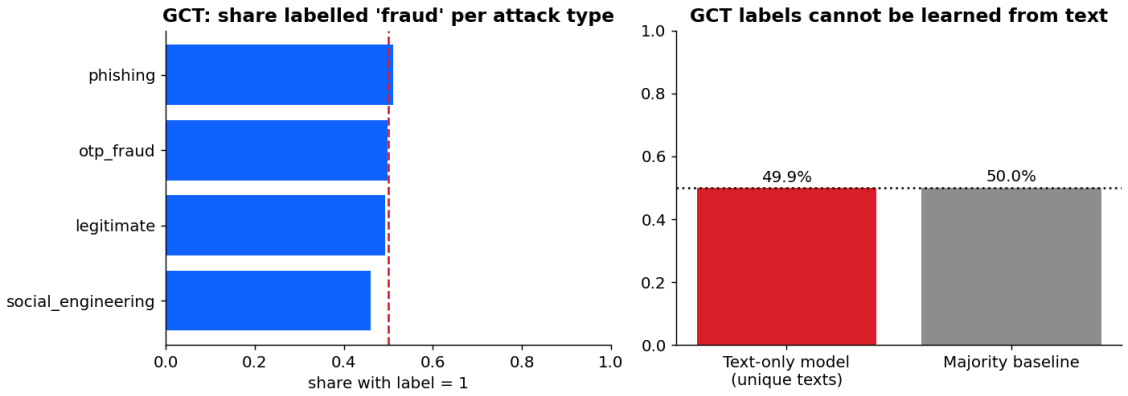


Figure 2. The GCT labels are statistically indistinguishable from random, so the dataset cannot support supervised training.

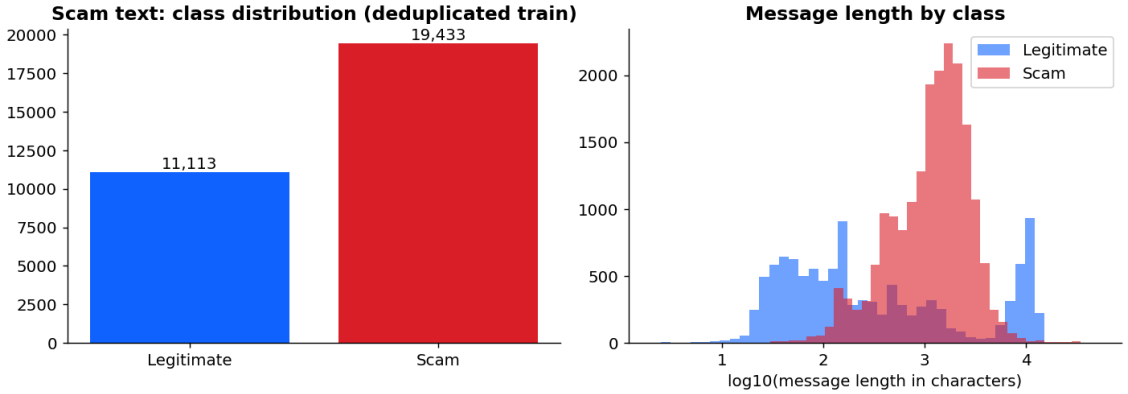


Figure 3. Scam-text class balance and message-length distribution after deduplication.

9. Machine Learning Methodology and Results

Pipeline. Raw data -> audit -> cleaning (duplicate and conflict removal, leak removal, text normalisation) -> feature extraction -> model comparison on validation data -> refit on all training data -> single evaluation on a held-out test set -> model saved with joblib. Random seeds are fixed (42) and the full pipeline is reproducible with `python train.py`.

Models compared. Logistic Regression, Naive Bayes, Linear SVM (probability-calibrated), Decision Tree, Random Forest (and Gradient Boosting for URLs). Metrics: accuracy, precision, recall, F1, ROC-AUC, PR-AUC and confusion matrix. Because the e-mail data are about 0.5-3% positive, PR-AUC is reported alongside accuracy.

9.1 Scam-text module

Model (validation split)	F1	ROC-AUC
Logistic Regression	0.9937	0.9994
Naive Bayes	0.9789	0.9957
Linear SVM	0.9956	0.9995
Decision Tree	0.9802	0.9711
Random Forest	0.9903	0.9985

Table 2. Validation comparison; the calibrated Linear SVM was selected.

On the held-out, leak-free test set (n=5,949; 4,482 scams) the selected model obtained accuracy 0.9971, precision 0.9991, recall 0.9971, F1 0.9981, ROC-AUC 0.9997 with 4 false positives and 13 false negatives. Because scam transcripts are much longer than genuine messages, a trivial length-only classifier reaches 85%; accuracy was therefore also measured by length bucket: <200 chars (SMS-like): 98.8% (n=877); 200-1000: 99.7% (n=1933); >1000 (long dialogue): 100.0% (n=3139). Even so,

templated scripts mean near-duplicates may remain between train and test, so real-world accuracy on unseen scam styles is likely lower than the headline figure.

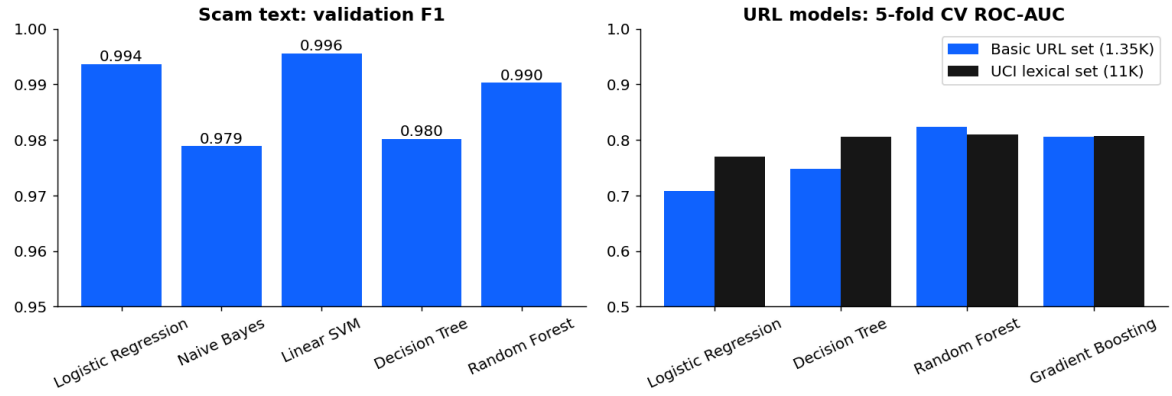


Figure 4. Left: validation F1 of the text models. Right: cross-validated ROC-AUC of the URL models.

9.2 URL module

Dataset	Unique rows	Selected model	Hold-out ROC-AUC	F1	Recall
Basic lexical set	1,354	Random Forest	0.814	0.586	0.589
UCI lexical set	11,055	Random Forest	0.827	0.739	0.841

Table 3. URL models (80/20 stratified hold-out; model chosen by 5-fold cross-validation).

These models are screening aids, not verdicts: they see only the URL string, not page content, domain age or reputation. The app therefore always shows red flags next to the score.

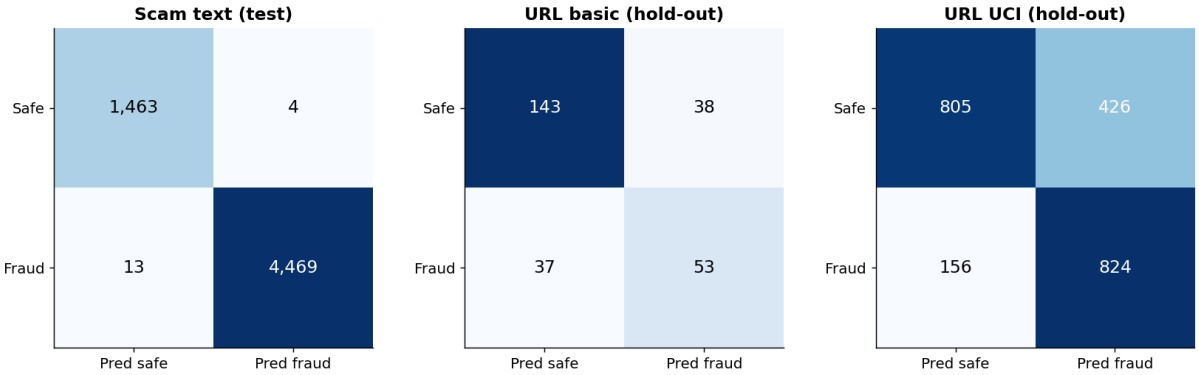


Figure 5. Confusion matrices on held-out data.

9.3 E-mail module and the domain-shift finding

The e-mail text model (TF-IDF + Linear SVM) scores F1 0.88, ROC-AUC 0.987 on a held-out part of the Enron sample. On the independent 84-e-mail set it drops to ROC-AUC 0.49, i.e. chance level, and the scam-text model and simple structural rules also fail there. The explanation is that the negatives are all Enron-era corporate mail, so the model partly learns writing style. This is a negative but valuable result: it shows why held-out accuracy from a single corpus can mislead. The structural-feature model on the 525K table reaches only PR-AUC 0.23 (positive rate 3.0%) and is not used in the app. The e-mail page is labelled experimental.

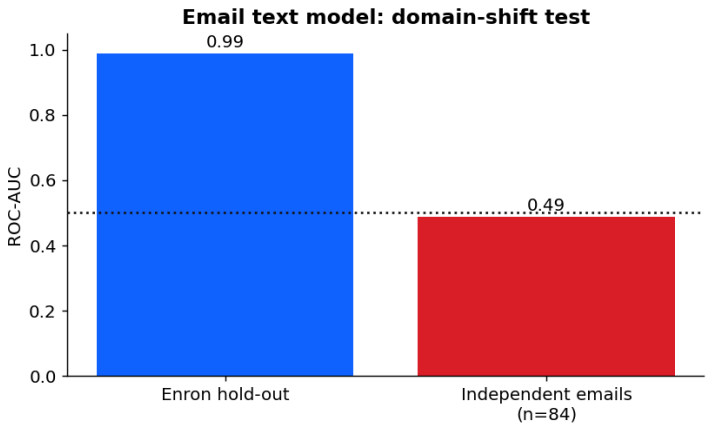


Figure 6. Held-out vs independent e-mail performance.

10. System Architecture and User Interface

The app (Streamlit) has pages for Home, Scam Message, URL Check, E-mail Check, Threat Dashboard, Model Report and About. The user pastes text or a URL and presses a button; the app displays the verdict (Legitimate / Suspicious / Fraudulent), the fraud probability, the model confidence and a 'Why?' panel. Trained models (about 11 MB) are stored in the repository so the application deploys without retraining. Unit tests cover text cleaning, URL features, e-mail features and the end-to-end predictors.

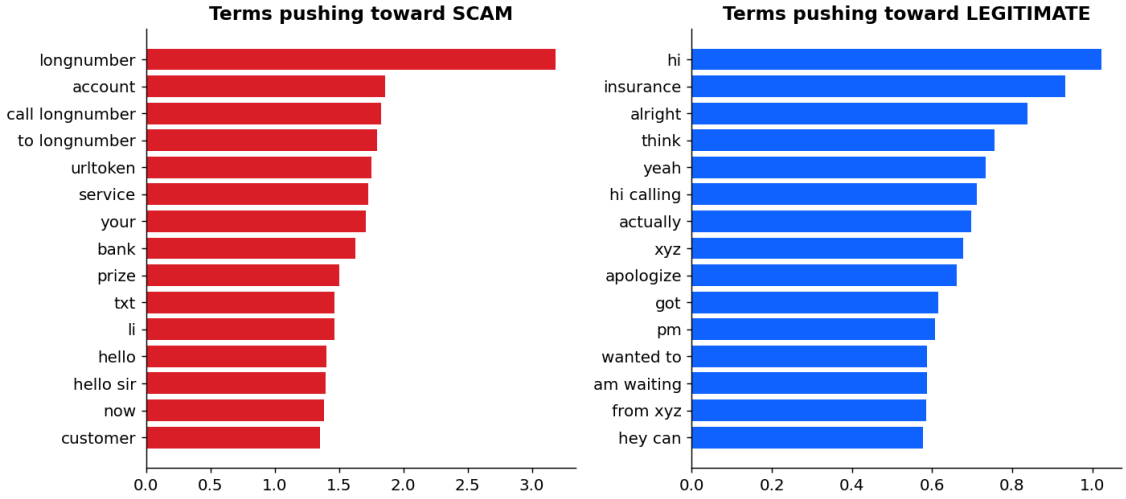


Figure 7. Terms with the largest positive and negative weights in the scam-text model.

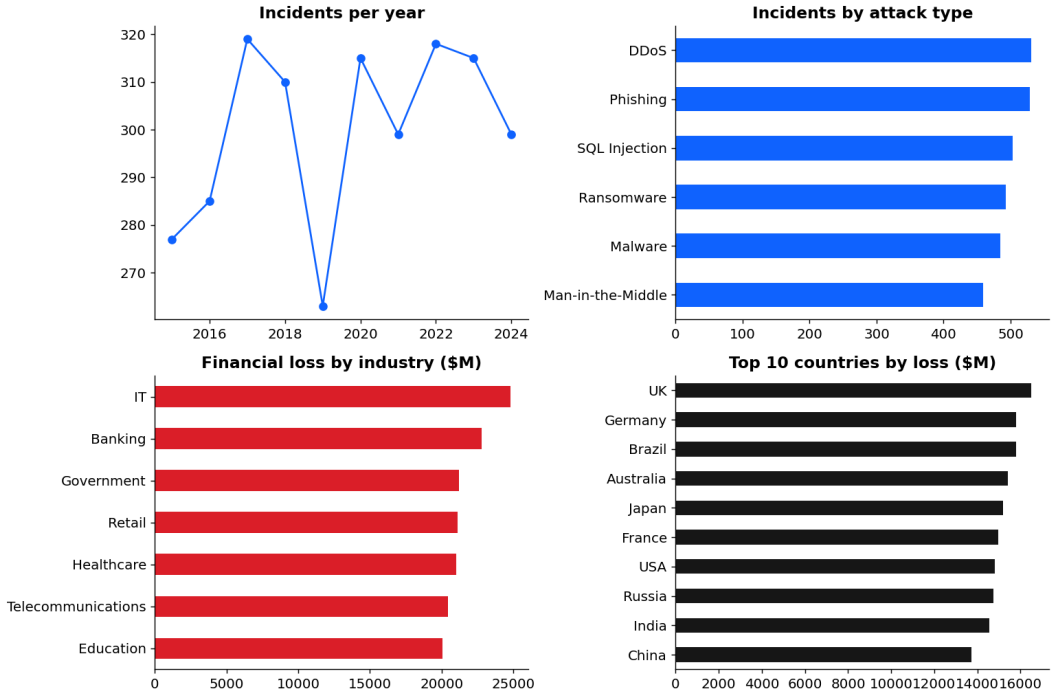


Figure 8. Dashboard charts from the Global Cybersecurity Threats dataset (3,000 incidents, 2015-2024).

11. Expected Outcomes and Achieved Results

- Working explainable detector for scam text (hold-out F1 0.998) and phishing URLs (ROC-AUC 0.81-0.83).
- A reproducible data-audit report documenting five dataset problems and how each was handled.
- A deployable Streamlit application, GitHub-ready repository, unit tests, Lean Canvas, presentation and screenshots.
- A documented negative result (e-mail domain shift) that guides future data collection.

12. Limitations

- No labelled corpus of short Indian SMS scams was supplied; short-message accuracy comes from the short items in the transcript set plus keyword rules.
- URL models use lexical features only; they can flag legitimate long or hyphenated URLs and miss well-crafted look-alike domains.
- The e-mail model does not generalise beyond its training corpus.
- English only; no adversarial testing; probabilities are model scores, not guarantees.

13. SDG Alignment

SDG 9 - Industry, Innovation and Infrastructure: AI Shield applies machine learning to make digital-payment infrastructure safer (target 9.c / 9.b: innovation and resilient digital infrastructure). **SDG 16 - Peace, Justice and Strong Institutions:** by helping users recognise cyber-enabled financial crime and pointing them to official reporting channels, the project supports target 16.4 (reducing illicit financial flows and organised crime) and trust in institutions.

14. Future Scope

- Train on a labelled SMS corpus (e.g., UCI SMS Spam Collection) and locally collected Indian scam messages, including Hindi and Gujarati.
- Use real phishing e-mail corpora and add domain-shift tests as a standard step.

- Add URL reputation, WHOIS domain age and TLS features; add QR-code scam detection.
- Browser extension, Android share-sheet integration and a WhatsApp bot; a user-feedback loop for continuous retraining.
- Compare transformer models (e.g., multilingual BERT) against the TF-IDF baseline.

15. Conclusion

AI Shield shows how applied machine learning can help ordinary users recognise fraud while staying honest about what the models can and cannot do. The audit uncovered unusable labels, duplicates and leakage that would have inflated results; correcting them produced trustworthy numbers and a clearer picture of where more data is needed. The result is a working, explainable, deployable prototype aligned with SDG 9 and SDG 16, and a foundation that can be strengthened with better Indian SMS and e-mail data.

16. References

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[11] United Nations Sustainable Development Goals - https://sdgs.un.org/goals ; IBM SkillsBuild ML & Applied AI Internship resources (BharatCares, 2026).